

Deep Learning: Homework 1

May 11, 2021

1. Your task is to find a linear approximation of the function $\sqrt{1+x}$, $x \in [0, 1]$. Your homework should contain the following steps:

- a) Generate $N = 10000$ random numbers from $[0, 1]$:

$$x_1, x_2, \dots, x_N \in [0, 1],$$

and then obtain their labels: $y_i = \sqrt{1+x_i}$, $i = 1, 2, \dots, N$.

- b) Do linear regression on your generated data using the closed form solution.
 - c) Do linear regression on your generated data using the library *sklearn*.
 - d) Do linear regression on your generated data implementing the gradient descent algorithm by yourself.
 - e*) Do linear regression on your generated data using tensorflow.
 - f) Sketch the graphs of all approximations on one graph.
 - g) Compare all solutions with the first degree Taylor approximation of the function $\sqrt{1+x}$.
- 2*. a) How will you define polynomial regression inspired from linear regression?
 - b) Can you implement the polynomial regression using linear regression?